Understanding Employee Attrition: An Exploratory HR Analytics Study of the IBM HR Dataset Using Python

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1. Project Definition

When employees leave a company, it can lead to higher recruitment costs, affect team stability, and lower overall productivity, This project will analyze the "IBM HR Analytics Employee Attrition & Performance" dataset, a fictional sample created by IBM data scientists, to find out which employee characteristics are associated with attrition.

Problem Statement: Organizations collect a large amount of employee data, but identifying which departments or groups of employees have the highest turnover can still be challenging. This project asks which job, pay, workload, and satisfaction factors are associated with employees leaving, and how visual analysis can make these patterns easy for HR decision-makers to understand.

Scope: The project will focus on checking data quality, cleaning the dataset, analyzing employee attrition, creating visualizations, and performing basic statistical tests. If time permits, a predictive model (logistic regression) will also be built. The project will report associations, not causes, and its findings will describe this dataset only, not real IBM employees.

2. Objectives

1. Check and clean the dataset, identify data quality issues, and remove columns that do not provide useful information.
2. Calculate the overall employee attrition rate and check whether the dataset has an imbalance between employees who leave and those who stay.
3. Compare attrition rates across different job roles, departments, job levels, and age groups.
4. Analyze how factors such as overtime, salary, years of service, and employee satisfaction are related to attrition.
5. Study the relationships between numerical HR variables to identify any noticeable patterns.
6. Present the main findings and explain how they could help HR teams understand employee attrition and make informed decisions.
7. Optional: Develop a basic classification model to predict employee attrition and evaluate its performance using suitable metrics, such as precision, recall, F1-score, and ROC-AUC

3. Dataset and Dataset Use Case

- **Name:** IBM HR Analytics Employee Attrition & Performance
- **Source:** Kaggle (pavansubhasht/ibm-hr-analytics-attrition-dataset), originally IBM Watson Analytics sample data .
- **Nature of data:** Fictional/synthetic data created by IBM data scientists. It is not real employee data.
- **Structure:** 1,470 rows and 35 columns in the standard version, one row per fictional employee.
- **Target variable:** Attrition (Yes/No).
- **Key features:** OverTime, JobRole, Department, Age, MonthlyIncome, JobLevel, YearsAtCompany, JobSatisfaction, WorkLifeBalance, EnvironmentSatisfaction, BusinessTravel, DistanceFromHome.
- **Cleaning plan:** The data will first be checked for missing values and duplicates. Constant columns (EmployeeCount, StandardHours, Over18) and the identifier EmployeeNumber will be removed. DailyRate, HourlyRate, and MonthlyRate will be checked for any relationship with income and excluded if none is found. Coded ratings will be converted to readable labels, and age groups and a binary attrition flag will be created.
- **Use in HR decision-making:** The analysis will show where attrition is concentrated and which workload and satisfaction factors go along with it. In a real setting, this could help HR decide where to focus retention discussions. Because the data is fictional, this project demonstrates a method rather than drawing real-world conclusions.

4. Proposed Data Analysis

The analysis will use descriptive statistics, attrition-rate tables by group, crosstabs, correlation analysis, and outlier and distribution checks. Chi-square and Mann-Whitney U tests, along with

effect sizes, will be used to check whether the observed differences could be explained by chance alone.

Optional extension: A stratified 80/20 train-test split will be used, followed by logistic regression with class weighting inside a scikit-learn pipeline. The model will be evaluated using precision, recall, F1, and ROC-AUC, and compared against a majority-class baseline. All preprocessing will be fitted on the training data only, to avoid data leakage.

5. Proposed Visualizations and Expected Outcomes

All categorical charts will display the percentage of employees who left rather than just the total number of employees, along with the group size (n). The expected outcomes are only assumptions to be tested during the analysis, not confirmed results. The "Actual Result" column will be completed once the analysis is finished.

	Visualization	Question answered	Expected outcome (hypothesis)	Actual result
1	Attrition distribution (bar chart)	How common is attrition?	A minority of employees leave	To be completed after analysis
2	Attrition rate by OverTime (bar chart)	Does overtime relate to attrition?	Higher among overtime workers	To be completed after analysis
3	Attrition rate by Department and JobRole (horizontal bars)	Where is attrition concentrated?	Rates vary across roles; small groups flagged	To be completed after analysis
4	Attrition rate by age group (bar chart)	Does age matter?	Higher in younger age groups	To be completed after analysis
5	MonthlyIncome by Attrition (box plot)	Do leavers earn differently?	Lower median income for leavers	To be completed after analysis
6	Attrition rate by JobLevel (bar chart)	Is attrition linked to seniority?	Higher at junior levels	To be completed after analysis
7	Satisfaction measures vs attrition (3-panel bar chart)	Do less satisfied employees leave more?	Lower satisfaction, higher attrition	To be completed after analysis
8	YearsAtCompany by Attrition (box plot)	Does tenure relate to leaving?	Shorter tenure for leavers	To be completed after analysis
9	Correlation heatmap	Which variables move together?	Strong links among age, tenure, income, and job level	To be completed after analysis

	Visualization	Question answered	Expected outcome (hypothesis)	Actual result
10	Attrition rate by BusinessTravel (bar chart)	Does travel frequency relate to attrition?	Higher among employees who travel frequently	To be completed after analysis
11	DistanceFromHome by Attrition (box plot)	Does commute distance relate to leaving?	Leavers may live slightly farther away, but the difference may be small	To be completed after analysis
12	Attrition rate by OverTime × JobRole (heatmap)	Does the overtime pattern differ across roles?	The overtime gap may be larger in some roles; cells with few employees will be flagged	To be completed after analysis

6. Python Libraries and Tools

Tool	Role in the project
Python 3	Core programming language
Pandas	Loading, cleaning, grouping, crosstabs, attrition-rate tables
NumPy	Numeric operations, binning, array handling
Matplotlib	Base plotting, figure layout, exporting images
Seaborn	Statistical plots (box plots, heatmap, bar charts)
SciPy	Chi-square and Mann-Whitney U tests
Scikit-learn (optional)	Train/test split, pipeline, logistic regression, evaluation metrics
Jupyter Notebook / Google Colab	Interactive development and documentation
Git and GitHub	Version control, collaboration, and submission

7. Expected Results / Outcomes

- A well-documented notebook that clearly explains each step, from preparing the raw data to creating the final visualizations.
- A set of clear visualizations that answer the HR questions in Section 5.
- A summary of the factors most associated with attrition in this dataset, written with appropriate caution.
- A README documenting methods, results, and limitations.

- If time permits, a baseline prediction model, evaluated with metrics suited to imbalanced data and compared against a majority-class baseline.

8. GitHub Repository and Documentation Plan

Repository link: https://github.com/milannakum54/ibm-hr-attrition-analysis.git

Planned structure:

```
ibm-hr-attrition-analysis/
├── README.md
├── requirements.txt
├── .gitignore
├── data/ (raw/ and processed/)
├── docs/ (data_dictionary.md, proposal.pdf)
├── notebooks/
│   ├── 01_data_validation_cleaning.ipynb
│   ├── 02_eda_visualization.ipynb
│   ├── 03_statistical_checks.ipynb
│   └── 04_optional_ml_model.ipynb
├── src/ (helper functions)
└── visualizations/ (exported PNG files)
```

The README will include the project summary, dataset description (stating that it is fictional), objectives, repository structure, run instructions, libraries, key findings, limitations, team roles, and references.

9. Team Members

Member	Roll No.	Responsibilities
MILAN NAKUM	IU2441230688	Repository setup, data validation and cleaning, categorical visualizations, README results section
RUDRA PATEL	IU2441230688	Data dictionary, numeric visualizations and correlation analysis, statistical checks, optional ML notebook, requirements file

Both members will also contribute to the documentation.

10. Limitations and Timeline

Limitations: The data is synthetic, so the patterns reflect how it was built and not real workplaces. It is a single snapshot with no documented date or sampling method. Attrition is imbalanced, some groups are small, and results show association, not causation.

Timeline:

Week	Activity
Week 1	Repository setup, data validation, and cleaning
Week 2	Visualizations and attrition-rate analysis
Week 3	Statistical checks and optional model
Week 4	Documentation, README, and final submission

11. Conclusion

This project will use Python to explore a well-known HR dataset and identify patterns associated with employee attrition. Careful data validation, clear visualizations, and cautious interpretation should lead to an analysis that others can reproduce. The data is fictional, so the results will describe associations within this dataset only.

References

1. Kaggle. IBM HR Analytics Employee Attrition & Performance (user: pavansubhasht). https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset
2. IBM Watson Analytics Lab, original source of the sample data, as cited in the R modeldata package documentation (https://modeldata.tidymodels.org/reference/attrition.html). The original IBM page may no longer be available.